

Bioreactor Productivity and Media Cost Comparison for Different Intensified Cell Culture Processes

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DOI 10.1002/btpr.2415

Published online January 28, 2017 in Wiley Online Library (wileyonlinelibrary.com)

*Process intensification in biomanufacturing has attracted a great deal of interest in recent years. Manufacturing platform improvements leading to higher cell density and bioreactor productivity have been pursued. Here we evaluated a variety of intensified mammalian cell culture processes for producing monoclonal antibodies. Cell culture operational modes including fed-batch (normal seeding density or high seeding density with N-1 perfusion), perfusion, and concentrated fed-batch (CFB) were assessed using the same media set with the same Chinese Hamster Ovary (CHO) cell line. Limited media modification was done to quickly fit the media set to different operational modes. Perfusion and CFB processes were developed using an alternating tangential flow filtration device. Independent of the operational modes, comparable cell specific productivity (fed-batch: 29.4 pg/cell/day; fed-batch with N-1 perfusion: 32.0 pg/cell/day; perfusion: 31.0 pg/cell/day; CFB: 20.1 – 45.1 pg/cell/day) was reached with similar media conditions. Continuous media exchange enabled much higher bioreactor productivity in the perfusion (up to 2.29 g/L/day) and CFB processes (up to 2.04 g/L/day), compared with that in the fed-batch processes (ranging from 0.39 to 0.49 g/L/day), largely due to the higher cell density maintained. Furthermore, media cost per gram of antibody produced from perfusion was found to be highly comparable with that from fed-batch; and the media cost for CFB was the highest due to the short batch duration. Our experimental data supports the argument that media cost for perfusion process could be even lower than that in a fed-batch process, as long as sufficient bioreactor productivity is achieved. © 2016 American Institute of Chemical Engineers *Biotechnol. Prog.*, 33:867–878, 2017*

Keywords: upstream processing, cost of goods, fed-batch, process intensification, perfusion

Introduction

A variety of cell culture processes could be used for producing recombinant proteins and monoclonal antibodies.^{1,2} Fed-batch processes have been widely adopted for clinical and commercial production due to their simplicity and scalability. Currently, the majority of the manufacturing capacity is built for fed-batch processes. Recent development of fed-batch processes has led to high peak cell density in the range of $20 - 30 \times 10^6$ cells/mL and high titer levels at >10 g/L in 18 days (bioreactor productivity at $0.5 - 0.8$ g/L/day).³ Perfusion processes are traditionally used to produce labile products such as blood clotting factors^{4–6} and enzyme products,⁷ though they have also been used to produce monoclonal

antibody products such as Remicade® (infliximab)⁸ and others.^{9,10} In perfusion cultures, continuous media exchange is used to reduce the product residence time in the bioreactor and the perfusion rate could vary based on the specific product and/or process requirement.

In recent years, perfusion-based process intensification has gained a considerable amount of momentum in upstream processing, largely driven by the desire in cost and footprint reduction and facility flexibility.^{11,12} With development in cell line, media, and cell retention devices, high cell density and productivity have been achieved in modern perfusion processes.¹² This renders it an attractive option even for monoclonal antibody production. For example, steady state cell density at $50 - 60 \times 10^6$ cells/mL could be typically achieved⁷ and bioreactor productivity at as high as 4 g/L/day at 2 medium exchange volume per bioreactor volume per

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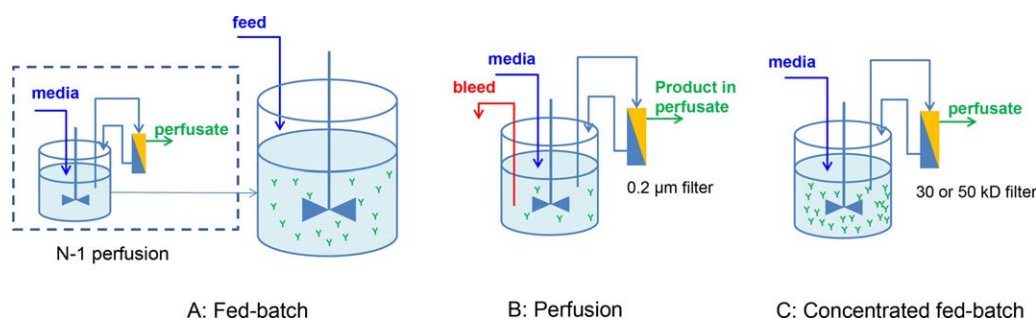


Figure 1. Illustration of different operational modes.

(A) Fed-batch (with or without N-1 perfusion), (B) Perfusion, (C) Concentrated fed-batch (CFB). ATF was used as the cell retention device in the perfusion and CFB processes.

day (vvd) has been recently reported for monoclonal antibody production.¹³ Very low cell specific perfusion rate (CSPR) at 15 pL/cell/day (medium exchange rate at 1 vvd) could also be achieved while maintaining bioreactor productivity at 1.9 g/L/day.¹⁴ Concentrated fed-batch (CFB) process also employs media exchange to maintain high cell density, while retaining product in the bioreactor. The difference between perfusion and CFB is primarily on the pore size of the hollow fiber membrane used (Figure 1). For antibodies, final titer at as high as 21.4 g/L could be achieved in 12 – 13 days using a Per.C6 cell line (peak density at $>150 \times 10^6$ cells/mL)¹⁵ or 25.3 g/L in 16 days with a Chinese Hamster Ovary (CHO) cell line (peak density at $>180 \times 10^6$ cells/mL).¹⁶ With the increase in bioreactor productivity, single-use facilities with smaller footprints and lower capital investment could be used to replace the large-scale stainless steel facilities (10,000 to 25,000 L) to produce the same amount of product with more frequent cycles or continuous processing.¹⁷

Though perfusion-based processes can generate and maintain high cell densities ($> 100 \times 10^6$ cells/mL) in the bioreactor using filtration-based cell retention devices, such as alternating tangential flow (ATF) and tangential flow filtration (TFF), high media exchange rates may be required to sustain such high cell density at acceptable viability.¹⁸ The cost of media associated with different processes is an important factor when evaluating which process can produce the same amount of product in a more cost-effective way. Even though media cost (\$/L) is modest, the impact of media on the overall manufacturing cost of goods (CoG) could become significant with high media exchange rates as the upstream operational expenses are closely related to media cost.¹⁹ Overall manufacturing CoG and the weight of upstream/downstream cost for producing a gram of product will change based on product titer and facility size.^{9,20–23} Among all the inputs for analyzing CoG, media usage and its cost is fixed once the process has been finalized, regardless of the change of facility, utility cost, etc. Additionally, media cost could be the dominating part of the upstream specific cost of goods for some processes.²⁴ One of the apparent goals for cell culture engineers is to reduce media cost while achieving high productivity. In this study, we demonstrate the development of different cell culture processes (fed-batch, perfusion, and CFB) with high bioreactor productivity using the same set of basal and feed media with minimum modifications. We further compare the bioreactor productivity from different operational modes and their associated

media cost. Estimated minimum required titer at different scenarios to make the media cost comparable between different processes is also discussed.

Materials and Methods

Cell line and inoculum expansion

A recombinant CHO cell line was used in the study. It was a glutamine synthetase (GS) cell line designed to produce a monoclonal antibody. The inoculum train started from vial thaw and expanded in shake flasks and Wave bioreactors (GE Healthcare, Uppsala, Sweden) at increasing volume. Both shake flasks and Wave bioreactors were maintained at 5% CO₂ and 36.5°C.

Bioreactor operations

The same 3 L bioreactors (Sartorius Stedim, Göttingen, Germany) with a single marine impeller and a drilled hole sparger (14 × 0.5 mm hole diameter) were used in all experiments. Proprietary basal²⁵ and feed media were used in the study. DO was controlled at 30% of air saturation using pure O₂. Temperature was controlled at 36.5°C. Agitation speed was controlled at 260 – 450 rpm. EX-CELL[®] antifoam (1% Simethicone Emulsion, Sigma-Aldrich, St. Louis, MO) was added in bolus or in a continuous fashion using a syringe pump (KDS-200, KD Scientific, Holliston, MA) for foam control.

For fed-batch cultures, bioreactors were inoculated at a target cell density of 0.5 or 2×10^6 cells/mL with an initial working volume of 1.5 L. Target inoculation density of 2×10^6 cells/mL was generated from a short-term perfusion for 3 days with a total of 3× media exchange using an ATF-2 unit (detailed hollow fiber filter information shown in Table 1) (Repligen Corporation, Waltham, MA). Bioreactors were fed according to glucose level on a daily basis within 14 days. The feed media are composed of Feed-A and Feed-B, which are rich in glucose, amino acids, vitamins, lipids, and trace elements.

For perfusion and CFB cultures, continuous media exchange was realized using an ATF-2 unit with different hollow fiber filter sizes (detailed hollow fiber filter information shown in Table 1). For CFB, a 50 kD PS filter was used; for perfusion, a 0.2 µm PES filter was used. Bioreactors were inoculated at a target cell density of 1×10^6 cells/mL with a working volume of 1.3 L. ATF flow rate of 0.8 LPM was used. Media exchange was typically started on day

Table 1. Main Characteristics of the Hollow Fiber Modules Used in the 3 L Bioreactors

	Length	Area	Fiber Number	Fiber Diameter	Pore Size	Material
Perfusion	61 cm	1300 cm ²	75	1 mm	0.2 μ m	Polyethersulfone (PES)
Concentrated fed-batch	61 cm	1400 cm ²	80	1 mm	50 kD	Polysulfone (PS)

2 and maximum exchange rate of 1 vvd was used. The maximum 1 vvd was used to reduce the media exchange volume required in the process. Two external pumps and scales were used. For perfusion cultures, cell bleed was started on day 8 to maintain desired cell density and viability. Additional 2.5 M D-Glucose solution (Sigma-Aldrich, St. Louis, MO) was fed continuously via a syringe pump (KDS-200) to control the glucose level at high cell densities.

Cell culture sample analysis

Samples from the 3 L bioreactors were taken daily and analyzed immediately. Viable cell density (VCD) and viability were measured using the trypan blue exclusion method on a Cedex Hi-Res cell counter (Roche Diagnostics GmbH, Mannheim, Germany). Offline pH, pO₂, and pCO₂ were measured using an ABL80 blood gas analyzer (Radiometer, Denmark). Glucose, lactate, glutamine, and glutamate were measured using a YSI Model 2700 analyzer (YSI Life Sciences, Yellow Springs, OH). Ammonium, electrolytes, and osmolality were measured using a Nova FLEX bioanalyzer (Nova Biomedical, Waltham, MA).

Supernatant titer was analyzed using an Agilent 1100 high-performance liquid chromatography (HPLC, Agilent Technologies, Santa Clara, CA) with a reversed phase column. The supernatant was separated from cell pellet by centrifuging the cell culture sample. Packed cell volume (PCV) was measured using a PCV cell counting tube (Sigma-Aldrich, St. Louis, MO). Unless specified, all the titer values referred to are supernatant titers. Monoclonal antibody in harvested cell culture fluids (HCCF) was purified on an Agilent 1100 HPLC system with a Protein A column (MabSelect SuRe™ resin, GE Healthcare, Uppsala, Sweden) or on a Tecan-based high-throughput method (Tecan Systems, San Jose, CA) with the same Protein A resin for quality attributes analysis. Aggregation level was determined on an Agilent 1100 HPLC system with a size exclusion column. Charge variants were determined on an Agilent 1200 HPLC system with an ion exchange column.

Productivity comparison and cost of goods analysis

PCV-adjusted titer was used to calculate the harvestable product as different process generated significantly different cell density/PCV. For example, a large portion of the effective bioreactor volume is composed of cells in the CFB processes (15–30% in our case, depending on the cell density and cell diameter). PCV-adjusted titer was calculated from the supernatant titer and PCV: PCV-adjusted titer = Supernatant titer $\times$ (1 – PCV). For the perfusion processes, the perfusate titer was used as the harvestable titer as the product from the culture fluid is directly collected after the 0.2 μ m hollow fiber filter.

The drug substance (DS) cost of goods analysis was performed using BioSolve Process economic modeling software (Biopharm Services Limited, Chesham, UK).

Results and Discussion

Cell culture performance in different operational modes

The same 3 L bioreactor configuration was used for all operational modes: fed-batch, perfusion, and CFB (Figure 1). ATF-2 was used in both perfusion and CFB processes. The 0.2 μ m filter used in the perfusion processes retains only cells in the bioreactor and product is collected from the perfusate side of the filter continuously. The 50 kD filter used in the CFB processes retains both cells and product in the bioreactor. The same set of basal and feed media (Feed-A and Feed-B) was used in all three operational modes since it provides a good foundation for meaningful cell/bioreactor productivity and media cost comparison. Spin-tubes were used for the rapid screening of the media combinations on cell growth, productivity, and product quality (data not shown).

Fed-batch mode. For the fed-batch mode, inoculation densities at 0.5 or 2 $\times 10^6$ cells/mL were used. The 2 $\times 10^6$ cells/mL inoculation density condition reduced the exponential growth phase for two days, thus peak density was reached on day 8, instead of day 10 in the 0.5 $\times 10^6$ cells/mL inoculation density condition. Peak cell density in the range of 20.2 – 26.2 $\times 10^6$ cells/mL was reached in both conditions (Figure 2A). Final viability was 92.5 $\pm$ 0.9% in the 0.5 $\times 10^6$ cells/mL inoculation density condition, and 82.4 $\pm$ 4.8% in the 2 $\times 10^6$ cells/mL inoculation density condition (Figure 2B). Limited antibody production was observed in the early exponential phase. Product titer increased at a consistent daily rate of 0.55 – 0.63 g/L in the late exponential to stationary phase for both conditions, demonstrating that the rate of production was comparable between the two. As a result, day 14 titer of 5.4 $\pm$ 0.1 g/L (qP of 29.4 $\pm$ 2.2 pg/cell/day) and 6.8 $\pm$ 0.2 g/L (qP of 32.0 $\pm$ 2.3 pg/cell/day) was reached respectively for the two inoculation densities (Figure 2C).

The bioreactor volumetric productivity (VPR) was calculated as the final bioreactor titer divided by the culture duration. The 2 $\times 10^6$ cells/mL inoculation density condition produced higher VPR than the 0.5 $\times 10^6$ cells/mL inoculation density condition (0.49 $\pm$ 0.01 g/L/day vs. 0.39 $\pm$ 0.01 g/L/day), largely because it reduced time in the initial growth phase and extended the production phase (Figure 2D). The high inoculation density condition received higher amount of feeds as feed volume was based on glucose level and linked to cell density. On average, 48% feed (percentage of the initial bioreactor volume) was used in the 0.5 $\times 10^6$ cells/mL inoculation density condition, and 52% feed was used in the 2 $\times 10^6$ cells/mL inoculation density condition. This may have contributed to the slightly higher qP in the 2 $\times 10^6$ cells/mL inoculation density condition.

Perfusion mode. In perfusion cultures, two different media scenarios were evaluated: one with only basal media, the other with basal and Feed-A. In both conditions, high viability at >85% was maintained throughout the whole culture duration when appropriate cell bleed was applied. In the basal only condition, average cell density at 44.0 $\pm$ 4.1 $\times 10^6$ cells/mL was maintained, with a daily productivity of

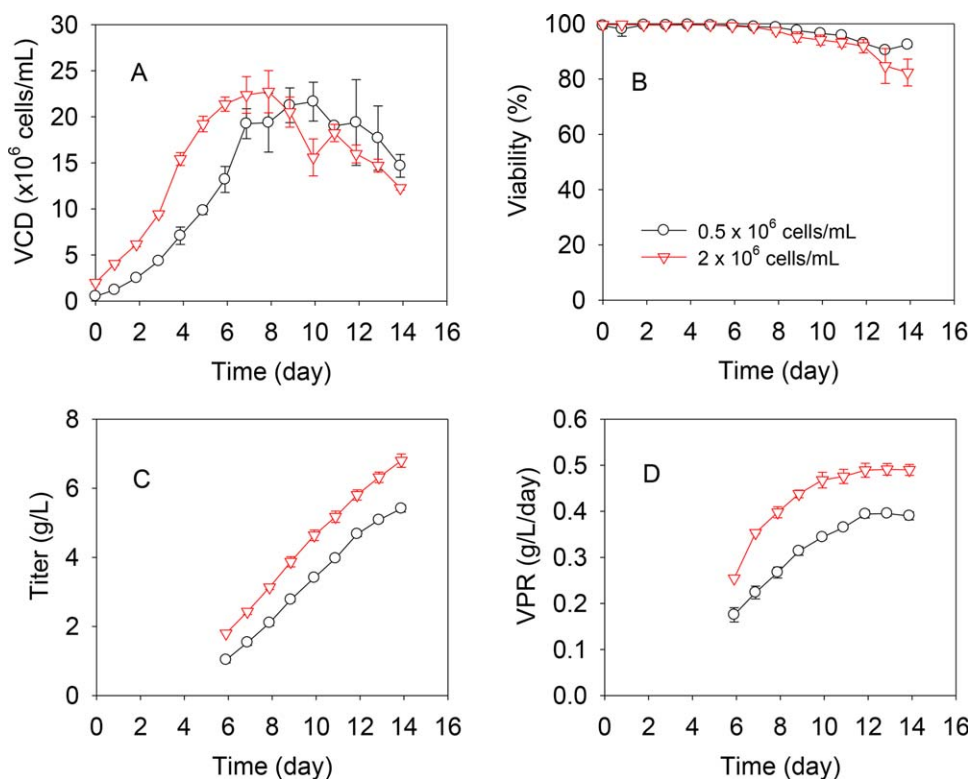


Figure 2. Culture performance in the fed-batch processes at 3 L scale.

(A) VCD, (B) viability, (C) titer, (D) volumetric productivity. Two inoculation densities were used: 0.5 and 2×10^6 cells/mL. Short-term perfusion was used to generate high final cell density in the N-1 step to support the inoculation density at 2×10^6 cells/mL. Error bar stands for one standard deviation (n = 3).

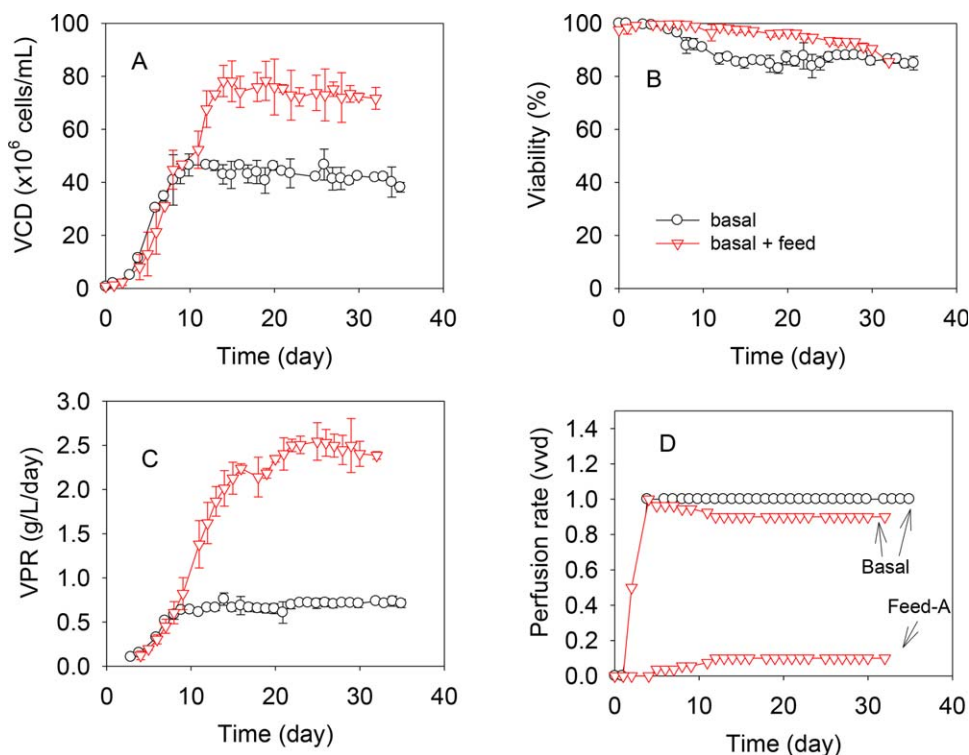


Figure 3. Culture performance in the perfusion processes at 3 L scale.

(A) VCD, (B) viability, (C) volumetric productivity (permeate titer), (D) Perfusion rate. Error bar stands for one standard deviation (n = 2).

0.70 ± 0.04 g/L/day from day 8 to day 32 (Figure 3A,C). In the perfusion cultures, the bioreactor volumetric productivity was calculated as the perfusate titer $\times$ perfusion rate since

the product is harvested directly from the perfusate side. The daily product transmission rates (perfusate titer/bioreactor titer) were still at $>80\%$ by the end of the perfusion cultures

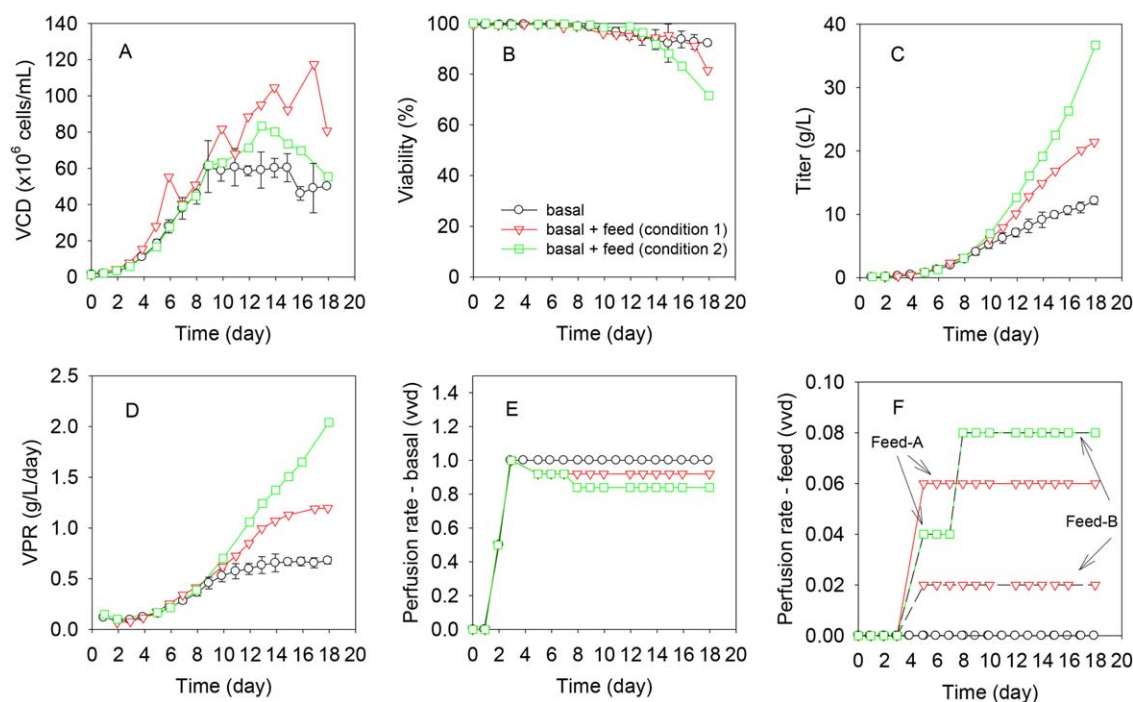


Figure 4. Culture performance in the CFB processes at 3 L scale.

(A) VCD, (B) viability, (C) titer, (D) Volumetric productivity, (E) Perfusion rate – basal, (F) Perfusion rate – feed. Error bar stands for one standard deviation ($n = 3$ for basal only condition, $n = 1$ for basal + feed conditions).

(data not shown). In the basal + feed condition, Feed-A was gradually introduced as part of the media exchange with the increase in cell density, while the total media exchange rate was kept at 1 vvd (Figure 3D). The amount of Feed-A in the total media exchange rate was increased until it reached 10% of the total media exchange on day 12 and was maintained at that level for the remaining of the culture duration. With the additional Feed-A, average cell density increased to $73.9 \pm 5.4 \times 10^6$ cells/mL and daily bioreactor volumetric productivity increased to 2.29 ± 0.28 g/L/day (Figure 3A,C). Compared with the basal only condition, the additional feed led to an increase in both steady state cell density and titer. Overall, the cell specific productivity was increased from 16.0 ± 1.2 pg/cell/day to 30.1 ± 2.3 pg/cell/day (minor product retention was omitted in calculating qP since daily transmission was high). As a result, a $\sim 230\%$ increase in bioreactor productivity was achieved in the basal + feed condition.

The addition of Feed-B in the process was also investigated and it was found that both cell specific and bioreactor volumetric productivities could be further improved (data not shown). However, the addition of Feed-B created more foaming with significant loss of cells. Sustaining steady state cell density for long duration was challenging, hence it was not included in the comparison here.

Concentrated fed-batch mode (CFB). Similar to the perfusion processes, basal media only condition and basal + feed condition were evaluated. No cell bleed was required and a higher cell mass accumulation was observed compared to the perfusion process. Since both Feed-A and Feed-B were used in the fed-batch processes, they were also evaluated in the CFB processes. Though addition of Feed-B led to an unsteady state in perfusion processes, there is less concern using it in the CFB processes, as no steady state cell density was required.

Improved cell culture performance and volumetric productivity were achieved with additional feeds (Figure 4). In the basal only condition, peak cell density reached $72.0 \pm 9.6 \times 10^6$ cells/mL with day 18 supernatant titer of 12.2 ± 0.6 g/L. For the basal + feed conditions, two different feed volumes were evaluated in media exchange: condition 1 used 6% Feed-A and 2% Feed-B and condition 2 used 8% Feed-A and 8% Feed-B (media exchange rates and feed rates shown in Figure 4E,F). Condition 1 yielded a peak cell density of 117.4×10^6 cells/mL and day 18 supernatant titer of 21.4 g/L. Condition 2 led to a peak cell density of 83.4×10^6 cells/mL and day 18 supernatant titer of 36.7 g/L. Significant increase of cell specific productivity to 45.1 pg/cell/day was achieved when the amount of Feed-A and Feed-B was increased in condition 2. The qP obtained here was similar to the levels reported in high productivity fed-batch processes.³ A similar boost of productivity was also achieved in the perfusion cultures where Feed-B was used (data not shown). Though viability declined faster in the basal + feed conditions than in the basal only condition, the final viabilities were still maintained at $>70\%$. It was also noticed that ATF-based CFB process (50 kD) can produce a higher viable cell density than perfusion (0.2 μ m) under the exact same culture condition (e.g., basal only condition, Table 2), similar to previously reported.²⁶ The exact reason for the observed cell density difference is yet to be understood better.

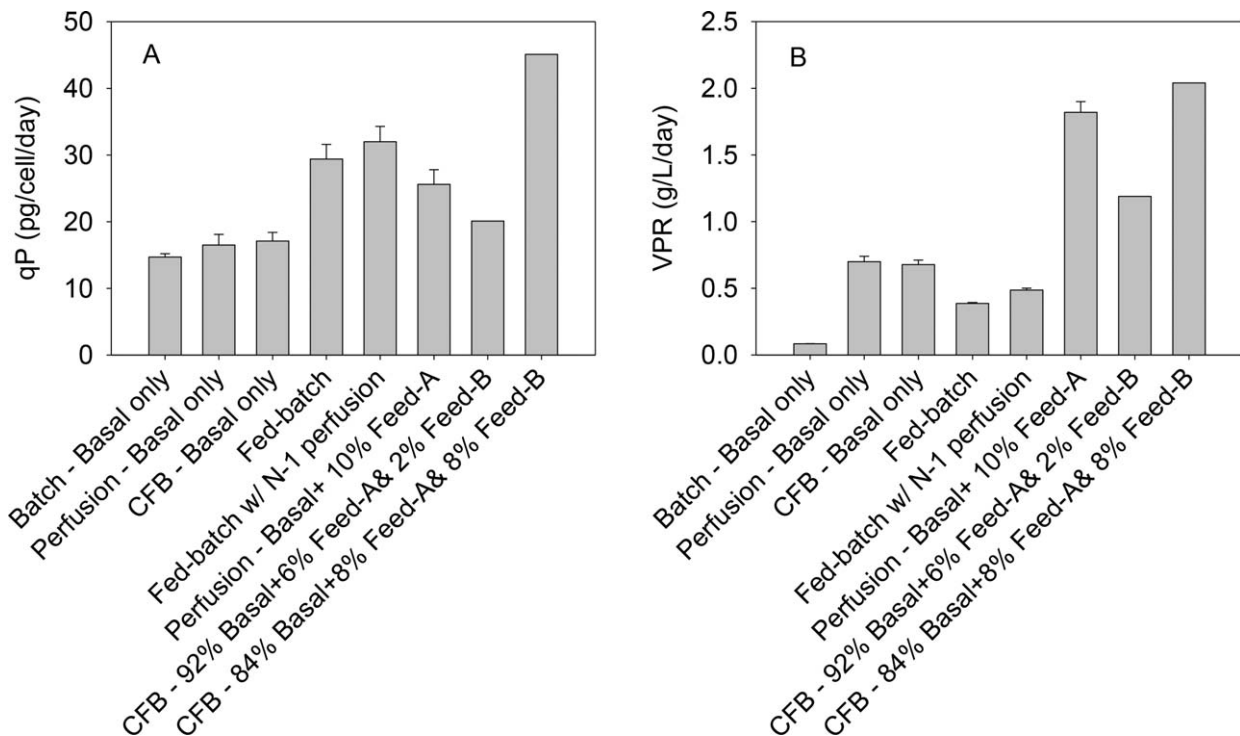
The two basal + feed conditions led to different levels of titer and productivity improvement. Cell specific productivity only increased modestly from the basal only condition (17.1 ± 1.3 pg/cell/day) to the basal + feed condition 1 (20.1 pg/cell/day) (Table 2). This could be partially due to the much higher cell density maintained in condition 1. The hypothesis is that nutrients were not sufficiently supplied for antibody production at that high density, thus the benefit of

Table 2. Summary of Cell Culture Performance Using Different Operational Modes (Average $\pm$ One Standard Deviation)

Operational Mode	Batch	Fed-Batch		Perfusion		CFB		
		Normal Fed-Batch	w/N-1 Perfusion	Basal Only	Basal+ Feed-A	Basal Only	Basal+ Feed Condition 1	Basal+ Feed Condition 2
Target N-stage production bioreactor total duration (day)	8	14	14	35	32	18	18	18
Seeding density ($\times 10^6$ cells/mL)	0.5	0.5	2	1	1	1	1	1
PVCD ($\times 10^6$ cells/mL)*	9.7 $\pm$ 0.2	22.1 $\pm$ 1.9	23.0 $\pm$ 2.1	42.9 $\pm$ 3.8	73.9 $\pm$ 5.4	72.0 $\pm$ 9.6	117.4	83.4
Final Viability (%)	98.7 $\pm$ 0.4	92.5 $\pm$ 0.9	82.4 $\pm$ 1.2	86.8 $\pm$ 2.6	85.5 $\pm$ 0.5	92.2 $\pm$ 0.9	81.5	71.5
Perfusion rate (vvd)	—	—	—	1	1	1	1	1
Cell bleed rate (vvd)	—	—	—	0.05 $\pm$ 0.01	0.06 $\pm$ 0.01	—	—	—
CSPR (pL/cell/day) [†]	—	—	—	23 $\pm$ 2	14 $\pm$ 1	—	—	—
Supernatant titer (perfusate titer for perfusion) (g/L)	0.7 $\pm$ 0.0	5.4 $\pm$ 0.1	6.8 $\pm$ 0.2	0.69 $\pm$ 0.05	2.29 $\pm$ 0.28	12.2 $\pm$ 0.6	21.4	36.7
qP (pg/cell/day)	14.7 $\pm$ 0.5	29.4 $\pm$ 2.2	32.0 $\pm$ 2.3	16.0 $\pm$ 1.2	31.0 $\pm$ 3.8	17.1 $\pm$ 1.3	20.1	45.1

*For perfusion, average cell density and perfusate titer from day 12 to end of culture were calculated.

[†]For perfusion cultures, average steady state CSPR was calculated over the course of the culture duration from day 12 to end of culture.

**Figure 5. Productivity comparison: (A) cell specific productivity and (B) volumetric productivity.**

Batch culture was included to compare the difference between batch and fed-batch culture productivity. Error bar stands for one standard deviation ($n = 2$ or 3).

high cell density on product formation was not maximized (supported by spent media analysis, data not shown). In basal + feed condition 2, further increase of Feed-A and Feed-B percentage in the media led to a significant increase in both cell specific productivity (45.1 pg/cell/day) and bioreactor volumetric productivity even though cell density was lower. This shows that how rich the media is for the specific cell line directly affects bioreactor productivity.

Cell specific productivity, bioreactor productivity, and product quality

To compare the difference of qP and VPR between processes, results from Figures 2–4 were extracted and presented in

Figure 5. Data from batch production with only basal media was also included for comparison. When only basal media was used, similar qP in the range of 14.7 – 17.1 pg/cell/day was reached in batch, perfusion, and CFB processes. In this case, the accumulated cell number directly impacts the product titer and volumetric productivity. As expected, batch culture had significantly lower VPR of only 0.08 g/L/day, while perfusion and CFB processes generated comparable VPR at 0.68 – 0.70 g/L/day due to the high cell density maintained.

Concentrated feeds are routinely used in the fed-batch processes for improving cell growth and cell specific productivity.^{3,27–29} In this study, the addition of feed media, Feed-A and Feed-B, contributed to a significant improvement in qP and VPR. For fed-batch cultures, qP was improved to

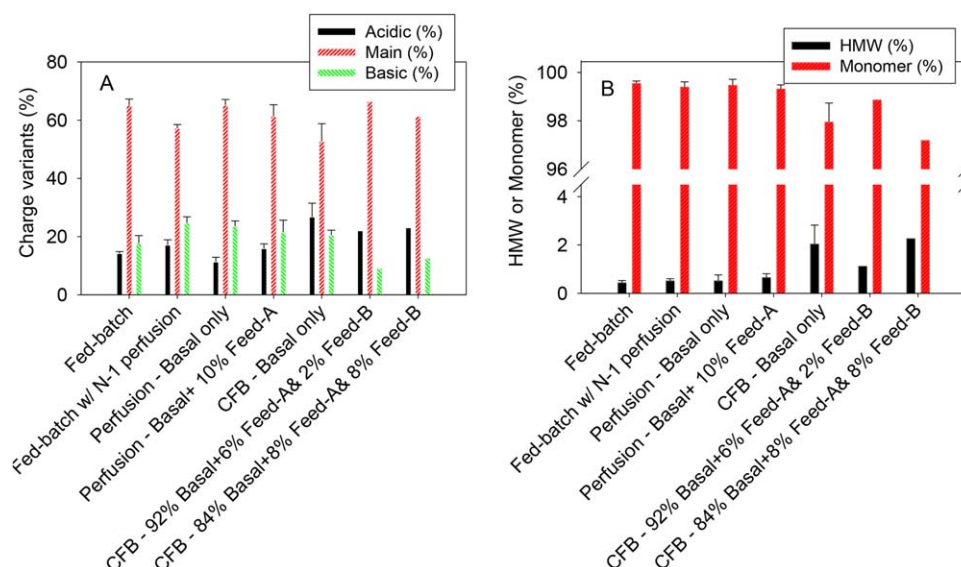


Figure 6. Product quality from different operational modes.

(A) Charge variants, (B) HMW and monomer. For fed-batch and CFB processes, final day bioreactor samples were analyzed for quality. For perfusion processes, day 12 to end of culture perfusate samples were analyzed for quality. Error bar stands for one standard deviation ($n = 2$ or 3).

29.4 – 32.0 pg/cell/day and VPR reached 0.39 g/L/day (0.5×10^6 cells/mL inoculation density) or 0.49 g/L/day (2×10^6 cells/mL inoculation density). High inoculation density from N-1 perfusion improved VPR since it reduced time in the growth phase and extended duration in the stationary phase where productivity is high.^{30,31} However, VPRs in the fed-batch cultures are still lower than those in the basal only conditions in the perfusion and CFB processes due to the significant cell density differences. Medium exchange with only basal medium at 1 vvd could easily generate cell densities 2 – 3 times higher than that for a fed-batch process (Table 2). Supplementation of 10% Feed-A in perfusion cultures led to ~230% higher VPR (2.29 g/L/day vs. 0.70 g/L/day) and ~90% higher qP (30.1 pg/cell/day vs. 16.0 pg/cell/day) compared with the basal only condition. Similarly, supplementation of different ratios of Feed-A and Feed-B improved VPR to 1.19 g/L/day (6% Feed-A + 2% Feed-B) and 2.04 g/L/day (8% Feed-A + 8% Feed-B) in the CFB processes. Increase in Feed-B amount was required to reach high cell specific productivity and volumetric productivity in the CFB process (Figure 4). Further improvement could still be made in cell specific and bioreactor productivities. Plasma cell secretion rates could be as high as 10,000 molecules/s, which equals to 8.6×10^8 molecules/day.^{32–34} This translates to the theoretical maximum secretion rate at up to 214 pg/cell/day for IgG antibodies (150 kDa). Recently, long-lived human plasma cells have been shown to be able to maintain IgG secretion rate at 120 pg/cell/day *in vitro*.³⁵ For genetically engineered mammalian cells, the maximum production rate has been estimated to be at ~100 pg/cell/day.³⁶ Improvement in qP could potentially come from the advancement in cell lines^{37,38} and media optimization.³ Theoretically, when the cell specific productivity is maximized at 100 pg/cell/day and a stable cell density is maintained at 100×10^6 cells/mL in a perfusion process, one could expect a daily productivity at as high as 10 g/L/day.

The product quality attributes from different operational modes were evaluated. Acceptable charge variants profiles and aggregation levels were obtained from all operational modes (Figure 6). CFB processes produced higher levels of

HMW and slightly higher acidic variants than the other two operational modes, largely due to the cell culture environment where product was exposed to. The product residence time in fed-batch and concentrated fed-batch cultures could be as long as the whole batch length, with the longest duration in the CFB processes. The highest HMW was from the basal only CFB process, implying that media composition could have played a role in HMW formation. Still, the HMW produced was less than 5% and the majority could be cleared by purification steps. For molecules that tend to aggregate, operational modes such as CFB will need to be carefully examined to ensure acceptable product quality attributes. For example, when the CFB process was extended to as long as 29 days, we observed an increase in the acidic variants and HMW levels from the basal only condition even though product titer was also increased substantially (Figure 7). On the other hand, even with the same high cell density environment and similar media compositions, perfusion cultures produced lower acidic variants and low HMW, likely due to the short residence time of the product in the bioreactor (1 vvd used, product continuously transported out of the bioreactor with a residence time of 3 days³⁹). The impact of operational modes will need to be assessed on a case-by-case basis for different molecules, as change in product quality is to be expected using different operational modes.¹⁶

Media cost analysis

As the change of cell line or media composition will greatly affect product titer/productivity, a comparison of different operational modes is only meaningful under the same cell line and media conditions. Here we compare the media cost from different operational modes directly using the cell culture performance obtained in small-scale bioreactors, assuming no significant productivity loss for different processes during scale-up. In our experience, only minimum and similar productivity loss was observed for scaling-up the fed-batch and perfusion processes (data not shown). We also assume that if a cell line/process has scale-up challenges, productivity loss will be observed for all the processes.

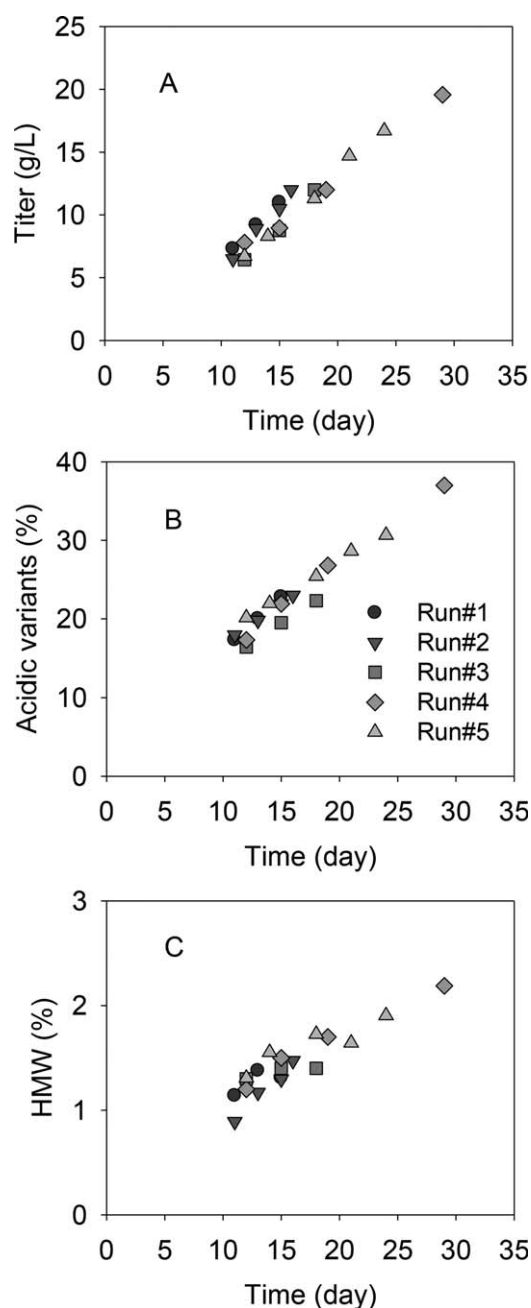


Figure 7. Time course of (A) Titer, (B) Acidic variants, and (C) HMW in the basal only CFB process with extended culture duration.

Thus, comparison using small-scale data could still serve as a good qualitative analysis for large-scale bioreactors. It has to be pointed out that the perfusion rates were not optimized in our study on a cell specific basis in the exponential growth phase. Instead, a fixed perfusion rate at 1 vvd was used throughout the culture duration, and only the rates of basal and feeds were proportionally adjusted according to the change of cell density (Figures 3 and 4). Fine-tuning of cell specific perfusion rate at different culture phases could lead to further reduction in media usage and cost. However, that was not pursued in this study.

Table 3 shows the media unit cost and estimated media usage at different manufacturing scales. The media unit price used was according to the most available information at the time of analysis. Figure 8 shows the comparison of media

cost per gram of antibody produced. Based on the titer results from the previous section, different manufacturing scales could be utilized to produce the same amount of product. For example, 2,000 L fed-batch, 500 L perfusion, and 500 L CFB processes could be used to meet the same level of production demand.

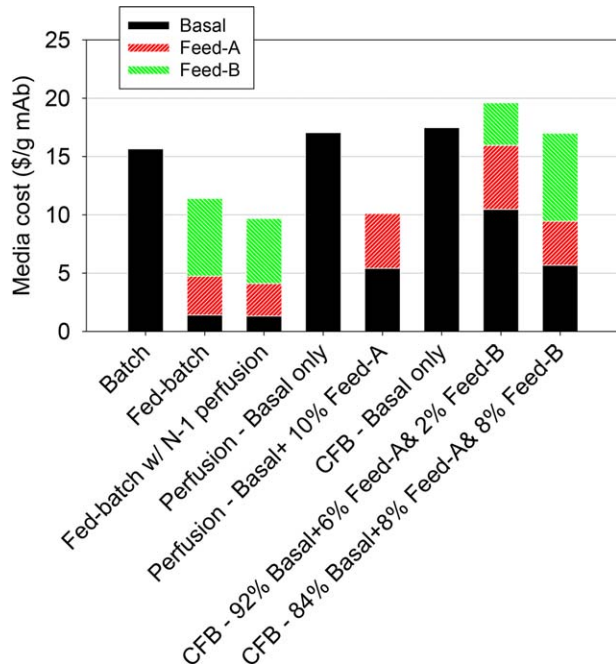
When only basal media was used, the media cost per gram antibody produced was high in both batch and perfusion processes. Reduction in media cost/g mAb was achieved when appropriate amount of feed media was used in the batch (thus turning into a fed-batch process) and perfusion processes. Even though feed media are more expensive than basal media, the increase in cell density and qP was more significant than the increase in media cost. The media cost for fed-batch with N-1 perfusion was lower than that from the normal fed-batch process, due to the fact that only $3 \times$ basal media exchange was required in N-1 perfusion for high inoculation density and the increase in titer outweighs the increase in media usage. The media cost for two conditions, fed-batch with N-1 perfusion and perfusion (basal + 10% Feed-A), was comparable at $\sim \$10/\text{g mAb}$. This shows that even though typically perfusion is regarded as more media intensive and has a higher media cost due to high perfusion rate, it could actually be competitive in terms of media cost as long as bioreactor productivity meets a certain threshold. In a recent analysis, 3 vvd perfusion rate was shown to be required to maintain cell density at 60×10^6 cells/mL in perfusion cultures and an improved process with 1 vvd to maintain the same cell density was projected to make the perfusion process favorable in terms of CoG.²⁴ This analysis correlates well with our experimental data and analysis.

Media cost for the CFB process did not follow the trends in other operational modes. In the basal only condition, the cost ($\$17.5/\text{g mAb}$) was comparable with those from the batch and perfusion processes. Different from the fed-batch and perfusion cases, the media cost actually increased when feed media were used in the CFB process. For example, the cost increased to $\$19.6/\text{g mAb}$ when feed condition 1 was used. In this particular case, the increase in bioreactor productivity is not sufficient to make up for the high cost associated with the feed media used. It was reflected by the marginal cell specific productivity increase from 17.1 ± 1.3 pg/cell/day to 20.1 pg/cell/day in the feed condition 1 (Table 2). With the further increase of feed volume in condition 2, both qP and VPR were significantly improved. As a result, the media cost was reduced to $\$17.0/\text{g mAb}$, which was only slightly lower than the media cost from the basal only condition of $\$17.5/\text{g mAb}$. High media cost associated with the CFB processes is largely due to the considerable time of cell growth required. Significant titer increase did not happen until day 10 in the culture where cell density reached peak level. Thus, even though the qP and daily productivity was high in the CFB process with feed condition 2, the media cost/g mAb was much higher than the $\sim \$10/\text{g mAb}$ from the fed-batch and perfusion processes. For the CFB process, one approach to reduce the media cost could be extending the batch length with improved longevity; however, that has to be carefully examined since longer residence time in the bioreactor will likely impact product quality attributes.^{40,41} In our case, when the CFB culture was extended to longer duration, increase in acidic variants and HMW have been observed (Figure 7). Further media optimization such as replacing expensive components used and improvement in

Table 3. Media Cost and Media Usage at Different Scales

Operational Mode	Batch Basal Only	Fed-Batch		Perfusion		CFB		
		Normal Fed-Batch	w/N-1 Perfusion	Basal Only	Basal+ Feed-A	Basal Only	Basal+ Feed Condition 1	Basal+ Feed Condition 2
Bioreactor scale/ working vol (L)	2,000	2,000	2,000	500	500	500	500	500
Target N-stage produc- tion bioreactor total duration (day)	8	14	14	35	35	18	18	18
Target production duration (day)	8	14	14	28	24	18	18	18
Target supernatant titer (perfusate titer for perfusion) (g/L)*	0.7	5.4	6.8	0.69	2.29	12.2	21.4	36.7
Packed cell volume (PCV) (%)	2	5	5	10	16	15	30	30
PCV adjusted titer (g/L)	0.7	5.1	6.5	–	–	9.6	15.0	25.7
Harvestable mAb from production bioreactor (g)	1,372	10,260	12,920	9,660	27,480	4,800	7,490	12,845
Harvest yield (%)	95	95	95	100	100	95	95	95
mAb from HCCF (g)	1,303	9,747	12,274	9,660	27,480	4,560	7,116	12,203
media exchange ($\times$ WV)	–	–	3	33	33	16	16	16
Basal (L)	2,040	1,378	1,618	16,250	16,250	7,970	7,450	6,930
Feed-A (L)	–	324	342	–	1,290	–	390	460
Feed-B (L)	–	324	342	–	–	–	130	460
Cost of Basal (\$/L)	10	10	10	10	10	10	10	10
Cost of Feed-A (\$/L)	–	100	100	–	100	–	100	100
Cost of Feed-B (\$/L)	–	200	200	–	–	–	200	200
Total media cost (\$)	20,400	111,081	118,816	164,700	277,700	79,700	139,500	207,300
Basal \$/g mAb	15.7	1.4	1.3	17.0	5.4	17.5	10.5	5.7
Feed-A \$/g mAb	–	3.3	2.8	–	4.7	–	5.5	3.8
Feed-B \$/g mAb	–	6.7	5.6	–	–	–	3.7	7.5
Overall media \$/g mAb	15.7	11.4	9.7	17.0	10.1	17.5	19.6	17.0

*Target titer based on the results from 3 L bioreactors.


Figure 8. Media cost comparison of the different operational modes.

titer is required to reduce the media cost in the CFB process. To provide some estimation on the titer level required to make media cost comparable with those from the fed-batch

and perfusion processes, different titer scenarios at different operational modes were evaluated (Figure 9). The media usage, cost, and culture duration were kept the same as shown in Table 3. One notes that to make media cost comparable, the CFB titer need to be ~ 10 times higher than that in the fed-batch process, and ~ 25 times higher than the daily permeate titer in the perfusion process. To compare with the results shown in this study, it is estimated that bioreactor supernatant titer of 60 g/L on day 18 is required to make the CFB process cost-competitive ($\sim \$10/\text{g}$) from the media usage point of view. This requires almost a doubling of the current bioreactor volumetric productivity, likely will have to come from the increase of total cell number and cell specific productivity.

Overall manufacturing CoG

Despite the media cost difference, it is worthwhile to mention that biologics manufacturing at smaller scales in single-use facilities employing processes such as perfusion and CFB processes could reduce the capital investment significantly, thus provide a more flexible manufacturing strategy and rapid scale-out instead of scale-up when demand has ramped up.^{19,42} The fixed costs associated with conventional stainless steel facilities will likely transition to a variable cost structure. Based on the case study shown here, the media cost for perfusion processes actually could be lower than that for the fed-batch processes. Furthermore, the overall cost of producing a gram of product should be considered when different manufacturing scales are used to produce the

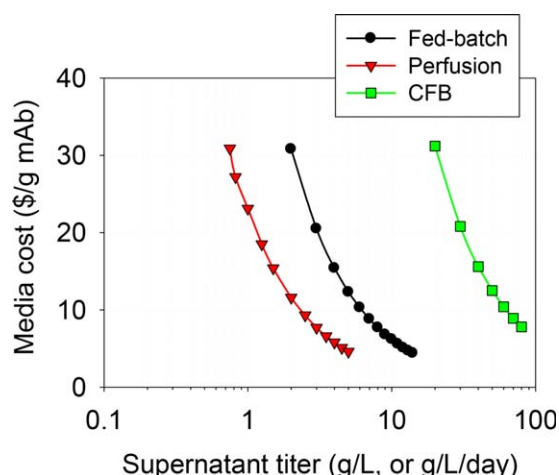


Figure 9. Sensitivity analysis of the media cost vs. supernatant titer in different operational modes.

For fed-batch and CFB, the titer unit is g/L; for perfusion the titer unit is g/L/day.

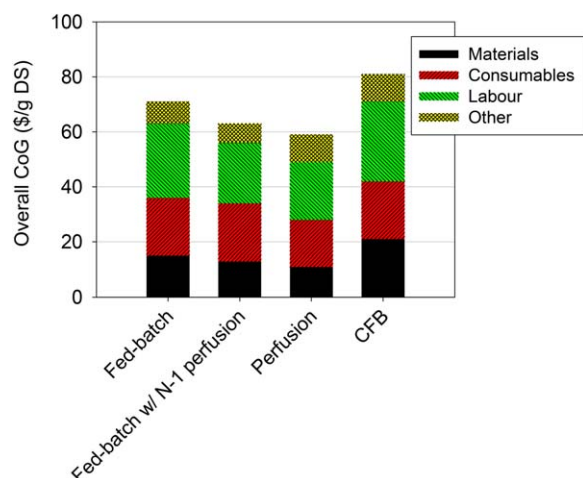


Figure 10. Operating costs/g drug substance for different upstream modes using the data from Table 3.

Materials include media, buffer, direct raw materials, WFI and PW, CIP, and QC tests. Consumables include bags, filters. Labor includes process, quality, and indirect. Other includes waste management, maintenance, utilities, and insurance.

same amount of product. This study provided an initial assessment of the productivity achievable converting one process to another, as well as the media cost associated with them. Development efforts such as media optimization and cell line improvement could be further directed according to the results shown here.

While significant, media cost is only one component making up the cost of goods for mAb drug substance production. Comparison of operating costs for drug substance was made under the different operational modes covered here using the BioSolve Process economic modeling software (Figure 10). It is a very basic comparison based on a single manufacturing train in a dedicated facility. Downstream operations are performed in the batch mode, even for the perfusion process, in order to keep the comparison consistent. For this exercise, disposable equipment costs were used and labor was kept consistent regardless of the operational mode. It is not

Table 4. Dedicated Facility Throughput (Single Manufacturing Train) for Different Upstream Modes Using the Data From Table 3

Operational Mode	Scale (L)	Batch/Year	Bioreactor Throughput/Year (kg)
Normal fed-batch	2,000	20	161
Fed-batch w/N-1 perfusion	2,000	20	202
Perfusion	500	8	204
CFB	500	15	151

expected to have significant operational manpower differences between different operational modes. The CoG/g DS are similar for fed-batch + N-1 perfusion (\$63/g) and perfusion (\$59/g) processes. The standard fed-batch and CFB processes run ~15% (\$71/g) and ~30% (\$81) higher in CoG/g DS. Though the assumptions for calculating the CoG could be different (annual output, scale, titer, etc.), reduction in the CoG has been observed over the years due to the significant improvement in titer and manufacturing processes.^{23,43} For example, the estimated CoG for a 1 g/L fed-batch process in 10,000 L production bioreactors was \$260/g in 2004,²⁰ and the estimated CoG for a 5 g/L fed-batch process in 5,000 L production bioreactor came down to an impressive \$39/g in 2013.⁹ Recent analysis also showed that the CoG for monoclonal antibody production could be in the range of \$57/g – \$95/g for different combinations of upstream (fed-batch, perfusion) and downstream (batch, continuous) operational modes.²⁴ Compared with some of the biological products price per unit, the drug substance CoG numbers covered here provide decent margin (0.4 – 4% of the sales price assuming a sales price range of \$2,000 – 20,000/g for antibody and Fc-fusion proteins).^{21,23,44} Though the drug substance CoG is a tiny portion, reducing the production cost is desired to improve the global accessibility of those therapies. Substantial effort is still needed to drive down the overall CoG to ~\$10/g.⁴⁵ Perfusion-based continuous processing has been shown to provide significant financial benefits in both monoclonal antibodies and less stable products.⁴⁶ It is worthwhile to note that the environmental footprint (such as water and energy consumption, waste generation, CO₂ emission, etc.)⁴⁷ and operational complexity/batch failure rates⁹ of different processes needs further comparison as that could potentially impact the final analysis.

Our study also shows that annual bioreactor throughput is ~33% higher using the fed-batch + N-1 perfusion and perfusion processes vs. using the fed-batch and CFB processes in this modeling scenario (~200 kg/year vs. ~150 kg/year, respectively, Table 4). It is estimated that a facility with 4 × 2,000 L fed-batch bioreactors (with N-1 perfusion) or 4 × 500 L perfusion bioreactors could deliver >500 kg drug substance/year assuming a downstream yield of >65%.

Conclusions

In this study, we compared the bioreactor productivity from different operational modes, including fed-batch, perfusion, and CFB processes. The same media set was easily adapted for different processes, providing a reasonable foundation for comparison. For the cell line studied, qP is highly dependent on the media used regardless of the operational modes, which leaves cumulative cell density as the main factor in determining product titer and bioreactor productivity. As a result, fed-batch cultures had the lowest bioreactor productivity (0.39 – 0.49 g/L/day) when compared with perfusion-

based cultures (2.29 g/L/day for perfusion and 1.19 – 2.04 g/L/day for CFB) due to the much lower cell density maintained. A significant advantage of perfusion is the ability to reach and maintain very high cell densities for product formation, which could offset the low qP in the basal only conditions. While the study was done with a single cell line, the impact of sequence/target integration on cell line performance will be explored in more detail for future work. Additionally, media optimization is further pursued for different operational modes since the presented work here was an initial assessment.

An often observed disadvantage of perfusion processes is the high media cost involved, since continuous media exchange is required to maintain the desired high viable cell density. Here we showed that media cost for high productivity perfusion cultures (2.29 ± 0.28 g/L/day) could actually be lower than that in the fed-batch process (with day 14 titer at 6.8 ± 0.2 g/L). This was partially made possible by the low cell specific perfusion rates (14 ± 1 pL/cell/day) used in the perfusion process. Media cost for the CFB process was the highest, even when a supernatant titer at as high as 36.7 g/L (PCV-adjusted titer at 25.7 g/L) was reached in 18 days. It is largely due to the fact that CFB cultures were not run as long as perfusion cultures, and a large portion of the culture duration was devoted to cell growth but not product formation. To further lower the media cost for the CFB process, it is suggested to fine-tune the medium exchange rate to have better medium utilization at the initial growth phase, or extend the batch duration, or further improve cell specific productivity via media optimization. In this study, we showed that operational modes could be switched from one to another fairly quickly using the existing media/cell line while maintaining desired productivity. As all operational modes were developed and assessed using the same media set for the same cell line, we envision that similar conclusions could be extended to other cell lines. It is important to note that achieving high productivity presents great opportunity for continuous bioprocessing utilizing perfusion technology even for stable products like monoclonal antibodies.

Acknowledgments

We would like to thank the Bioprocess Technical Operations group for media preparation and bioreactor operations, the In-Process Analytics and Purification groups for titer and quality analysis. We would also like to thank the constructive feedback from Nihal Tugcu, David Roush, and Kristin O'Neill.

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Manuscript received Aug. 30, 2016, and revision received Nov. 6, 2016.